

Linear Programming Course Summary

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1 Key Terms

- **Simplex Method:** An iterative algorithm used to solve linear programming problems by moving from one vertex of the feasible region to another until an optimal solution is found.
- **Slack Variable:** A non-negative variable added to a less-than-or-equal-to inequality constraint to transform it into an equality constraint.
- **Feasible Region:** The set of all points that satisfy all the constraints of a linear programming problem.
- **Basic Variable:** In the Simplex method, a variable that is expressed in terms of non-basic variables in a dictionary.
- **Non-basic Variable:** In the Simplex method, a variable set to zero in a given iteration of the algorithm.
- **Pivot Operation:** The process of exchanging a basic and a non-basic variable in the Simplex method to move to a new vertex of the feasible region.
- **Dictionary:** A representation of a linear programming problem's constraints and objective function where basic variables are expressed in terms of non-basic variables.
- **Auxiliary Problem:** A modified linear programming problem introduced to find an initial feasible solution when the origin is not feasible in the original problem.

2 Problem Types

- **Finding an Initial Feasible Solution:** The problem of finding a feasible solution to start the Simplex method, especially when the origin is not feasible. An auxiliary problem is introduced to find a feasible solution for the original problem.
- **Handling Infeasible Origins:** Addressing situations where the origin $(0,0)$ is not within the feasible region defined by the constraints of a linear programming problem. This requires a different starting point for the Simplex method.
- **Optimizing in Higher Dimensions:** Solving linear programming problems with more than two variables, requiring algebraic methods like the Simplex method rather than graphical solutions.
- **Using Dictionaries to Track Solutions:** Managing the basic and non-basic variables during the Simplex method iterations using a dictionary representation. The dictionary is updated with each pivot operation.

3 Algorithm Solutions

- **Simplex Method:** Iteratively move from one vertex of the feasible region to another by setting non-basic variables to zero and choosing a new set of (non)basic variables to improve the objective function value. This is equivalent to moving from vertex to vertex in the feasible region.
- **Graphical Method:** Identify the feasible region defined by the constraints. The optimal solution lies at one of the vertices of this region. Evaluate the objective function at each vertex to find the maximum or minimum value.
- **Pivot Operation:** Exchange one basic and one non-basic variable in the dictionary to move to a new vertex in the feasible region. This involves rewriting the equations to express the new basic variables in terms of the new non-basic variables.
- **Auxiliary Problem:** Introduce a new variable (x_0) and modify constraints to create a new problem where the origin is feasible. Solve this auxiliary problem. If the optimal solution has $x_0 = 0$, then this solution is feasible for the original problem.